

Memory Forensics

CSCI-4538-01

Professor Page

Lab 3: Maleficent

April 6, 2025

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Table of Contents

Executive Summary.....	3
Tools & Equipment.....	4
Procedure & Analysis.....	5
Challenges & Responses.....	22
Conclusion & Recommendations.....	23
References.....	24
Appendices.....	25

Executive Summary

This laboratory exercise focused on the analysis of a memory image to identify potential signs of malicious activity. Investigators utilized memory forensic tools to examine active and terminated processes, analyze memory regions, and extract suspicious executables for further review. The objective was to simulate a real-world investigation in which malware is detected and analyzed from volatile memory.

Through this process, the lab emphasized the importance of understanding process behavior, memory artifacts, and the steps necessary to uncover and examine potentially harmful software running on a compromised system.

Tools & Equipment

ITEM	Purpose	DATE USED
VMWare	Virtualization software to utilize Kali Linux.	04-01-25
Kali Linux VM	Operating System used for analysis	04-01-25
Memory Image	Title: <i>maleficent.vmem</i> at /home/relik/Downloads/maleficent.vmem)	04-01-25
Volatility 3	Tool used to conduct analysis on volatile memory.	04-01-25
Volatility 2	Tool used to conduct analysis on volatile memory.	04-01-25
VirusTotal	Analysis tool used to confirm presence of malware on image	04-01-25

Procedure & Analysis

Initially, investigators began by running the windows.info plugin to assess the system to be analyzing (Figure 1). What was found was that the system is a 32-bit Windows XP SP3 machine. The time the image was created was at approximately 2014-10-29 17:20:02 UTC.

```

1 Volatility 3 Framework 2.20.0
2
3 Variable      Value
4
5 Kernel Base   0x804d7000
6 DTB           0x32a000
7 Symbols       file:///home/relik/volatility3/volatility3/symbols/windows/ntkrnlpa.pdb/7E4571CB945F42D182C86ABE8E44D-1.json.xz
8 Is64Bit       False
9 IsPAE         True
10 layer_name    0 WindowsIntelPAE
11 memory_layer  1 FileLayer
12 KdDebuggerDataBlock 0x80545ce0
13 NTBuildLab    2600.xpsp_sp3_qfe.130704-0421
14 CSDVersion    3
15 KdVersionBlock 0x80545cb8
16 Major/Minor   15.2600
17 MachineType   332
18 KeNumberProcessors 1
19 SystemTime    2014-10-29 17:20:02+00:00
20 NtSystemRoot  C:\WINDOWS
21 NtProductType NtProductWinNt
22 NtMajorVersion 5
23 NtMinorVersion 1
24 PE MajorOperatingSystemVersion 5
25 PE MinorOperatingSystemVersion 1
26 PE Machine     332
27 PE TimeDateStamp Thu Jul 4 02:08:14 2013

```

Figure 1: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.info

Following the findings of the initial results, investigators then enumerated processes using the windows.pslist.PsList plugin to find all current processes on the system (Figure 2). Additionally, investigators also checked for hidden or terminated processes using the windows.psscan.PsScan plugin (Figure 3).

1 Volatility 3 Framework 2.20.0											
2											
3	PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4											
5	4	0	System	0x8a53b830	62	1095	N/A	False	N/A	Disabled	
6	596	4	smss.exe	0x8a31d020	3	19	N/A	False	2014-10-27 20:01:18.000000 UTC	N/A	Disabled
7	672	596	csrss.exe	0x8a177020	2	613	0	False	2014-10-27 20:01:20.000000 UTC	N/A	Disabled
8	696	596	winlogon.exe	0x89fb88d8	19	568	0	False	2014-10-27 20:01:20.000000 UTC	N/A	Disabled
9	740	696	services.exe	0x89fb7da0	15	307	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
10	752	696	lsass.exe	0x89f334b0	23	365	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
11	964	740	vmacthlp.exe	0x8a457410	1	25	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
12	976	740	svchost.exe	0x8a194880	16	193	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
13	1040	740	svchost.exe	0x8a484020	10	327	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
14	1180	740	svchost.exe	0x8a278b20	77	1591	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
15	1224	740	svchost.exe	0x8a16d980	6	87	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
16	1260	740	svchost.exe	0x8a309c10	13	175	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
17	1508	740	spoolsv.exe	0x8a0335b8	15	168	0	False	2014-10-27 20:01:23.000000 UTC	N/A	Disabled
18	1628	740	svchost.exe	0x8a227f18	4	108	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
19	1672	740	svchost.exe	0x8a16cda0	6	93	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
20	1684	740	CodeMeter.exe	0x8a078da0	6	126	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
21	1768	740	jqs.exe	0x89fd53b0	5	149	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
22	1800	740	mdm.exe	0x8a3766f0	5	85	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
23	1820	740	mysqld-nt.exe	0x8a029bd0	13	35325	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
24	1864	740	cygrunsrv.exe	0x8a170560	4	86	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
25	132	740	TNSLSNR.EXE	0x8a454980	3	89	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
26	184	1864	cygrunsrv.exe	0x8a076b20	0	-	0	False	2014-10-27 20:01:27.000000 UTC	2014-10-27 20:01:27.000000 UTC	Disabled
27	200	184	sshd.exe	0x8a362da0	3	86	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
28	192	740	extjob.exe	0x8a194da0	2	33	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
29	224	740	oracle.exe	0x8a039c10	19	453	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
30	288	740	vmtoolsd.exe	0x89f517c8	7	320	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
31	1272	740	alg.exe	0x8998e870	6	109	0	False	2014-10-27 20:01:34.000000 UTC	N/A	Disabled
32	1576	304	explorer.exe	0x89f506e0	19	1216	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
33	1176	1576	GrooveMonitor.exe	0x8a18c368	1	119	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
34	1232	1576	jusched.exe	0x899212d8	2	222	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
35	948	1576	rundll32.exe	0x8a2c9940	4	76	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
36	816	1576	vmtoolsd.exe	0x899a64f8	6	281	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
37	1988	1576	ctfmon.exe	0x8991d528	1	77	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
38	1140	1576	CodeMeterCC.exe	0x8991c8d8	1	82	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
39	1172	1576	MagicDisc.exe	0x89966da0	1	32	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
40	2360	1576	chrome.exe	0x899b3020	35	680	0	False	2014-10-27 21:59:40.000000 UTC	N/A	Disabled
41	2828	2360	chrome.exe	0x8993a7c8	5	135	0	False	2014-10-27 21:59:52.000000 UTC	N/A	Disabled
42	488	1576	cmd.exe	0x8969cb20	1	33	0	False	2014-10-29 16:52:05.000000 UTC	N/A	Disabled
43	2308	488	keylogger-local	0x8969d638	1	24	0	False	2014-10-29 16:52:10.000000 UTC	N/A	Disabled
44	1604	1576	cmd.exe	0x896a5020	1	32	0	False	2014-10-29 16:52:17.000000 UTC	N/A	Disabled
45	1456	1576	WINWORD.EXE	0x8a36bda0	9	920	0	False	2014-10-29 17:14:42.000000 UTC	N/A	Disabled
46	3148	2360	chrome.exe	0x898822d0	6	135	0	False	2014-10-29 17:18:15.000000 UTC	N/A	Disabled
47	3076	1576	ieplora.exe	0x89fbada0	16	413	0	False	2014-10-29 17:18:24.000000 UTC	N/A	Disabled
48	3852	3076	ieplora.exe	0x89658020	33	826	0	False	2014-10-29 17:18:25.000000 UTC	N/A	Disabled
49	360	288	cmd.exe	0x8a316470	0	-	0	False	2014-10-29 17:20:02.000000 UTC	2014-10-29 17:20:02.000000 UTC	Disabled

Figure 2: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.pslist.PsList

1 Volatility 3 Framework 2.20.0											
2											
3	PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4											
5	3852	3076	ieplora.exe	0x9658020	33	826	0	False	2014-10-29 17:18:25.000000 UTC	N/A	Disabled
6	488	1576	cmd.exe	0x969cb20	1	33	0	False	2014-10-29 16:52:05.000000 UTC	N/A	Disabled
7	2308	488	keylogger-local	0x969d638	1	24	0	False	2014-10-29 16:52:10.000000 UTC	N/A	Disabled
8	1604	1576	cmd.exe	0x96a5020	1	32	0	False	2014-10-29 16:52:17.000000 UTC	N/A	Disabled
9	3100	360	ipconfig.exe	0x9866afa0	0	-	0	False	2014-10-29 17:20:02.000000 UTC	2014-10-29 17:20:02.000000 UTC	Disabled
10	3148	2360	chrome.exe	0x98822d0	6	135	0	False	2014-10-29 17:18:15.000000 UTC	N/A	Disabled
11	1140	1576	CodeMeterCC.exe	0x991c8d8	1	82	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
12	1988	1576	ctfmon.exe	0x991d528	1	77	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
13	1232	1576	jusched.exe	0x99212d8	2	222	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
14	2828	2360	chrome.exe	0x993a7c8	5	135	0	False	2014-10-27 21:59:52.000000 UTC	N/A	Disabled
15	1172	1576	MagicDisc.exe	0x9966da0	1	32	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
16	1272	740	alg.exe	0x998e870	6	109	0	False	2014-10-27 20:01:34.000000 UTC	N/A	Disabled
17	816	1576	vmtoolsd.exe	0x99a64f8	6	281	0	False	2014-10-27 21:59:29.000000 UTC	N/A	Disabled
18	2360	1576	chrome.exe	0x99b3020	35	680	0	False	2014-10-27 21:59:40.000000 UTC	N/A	Disabled
19	752	696	lsass.exe	0x9f334b0	23	365	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
20	1576	304	explorer.exe	0x9f506e0	19	1216	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
21	288	740	vmtoolsd.exe	0x89f517c8	7	320	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
22	740	696	services.exe	0x89fb7da0	15	307	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
23	696	596	winlogon.exe	0x89fb88d8	19	568	0	False	2014-10-27 20:01:20.000000 UTC	N/A	Disabled
24	3076	1576	ieplora.exe	0x9fbada0	16	413	0	False	2014-10-29 17:18:24.000000 UTC	N/A	Disabled
25	1768	740	jqs.exe	0x9fd53b0	5	149	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
26	1820	740	mysqld-nt.exe	0xa029bd0	13	35325	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
27	1508	740	spoolsv.exe	0xa0335b8	15	168	0	False	2014-10-27 20:01:23.000000 UTC	N/A	Disabled
28	224	740	oracle.exe	0xa039c10	19	453	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
29	184	1864	cygrunsrv.exe	0xa076b20	0	-	0	False	2014-10-27 20:01:27.000000 UTC	2014-10-27 20:01:27.000000 UTC	Disabled
30	1684	740	CodeMeter.exe	0xa078da0	6	126	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
31	1672	740	svchost.exe	0xa16cda0	6	93	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
32	1224	740	svchost.exe	0xa16d980	6	87	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
33	1864	740	cygrunsrv.exe	0xa170560	4	86	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
34	672	596	csrss.exe	0xa177020	12	613	0	False	2014-10-27 20:01:20.000000 UTC	N/A	Disabled
35	1176	1576	GrooveMonitor.exe	0xa18c368	1	119	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
36	976	740	svchost.exe	0xa194880	16	193	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
37	192	740	extjob.exe	0xa194da0	2	33	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
38	1628	740	svchost.exe	0xa227f18	4	108	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
39	1180	740	svchost.exe	0xa278b20	77	1591	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
40	948	1576	rundll32.exe	0xa2c9940	4	76	0	False	2014-10-27 21:59:28.000000 UTC	N/A	Disabled
41	1260	740	svchost.exe	0xa309c10	13	175	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
42	360	288	cmd.exe	0xa316470	0	-	0	False	2014-10-29 17:20:02.000000 UTC	2014-10-29 17:20:02.000000 UTC	Disabled
43	596	4	smss.exe	0xa31d020	3	19	N/A	False	2014-10-27 20:01:18.000000 UTC	N/A	Disabled
44	200	184	sshd.exe	0xa362da0	3	86	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
45	1456	1576	WINWORD.EXE	0xa36bda0	9	920	0	False	2014-10-29 17:14:42.000000 UTC	N/A	Disabled
46	1800	740	mdm.exe	0xa3766f0	5	85	0	False	2014-10-27 20:01:26.000000 UTC	N/A	Disabled
47	132	740	TNSLSNR.EXE	0xa454980	3	89	0	False	2014-10-27 20:01:27.000000 UTC	N/A	Disabled
48	964	740	vmacthlp.exe	0xa457410	1	25	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
49	1040	740	svchost.exe	0xa484020	10	327	0	False	2014-10-27 20:01:21.000000 UTC	N/A	Disabled
50	4	0	System	0xa53b830	62	1095	N/A	False	N/A	Disabled	

Figure 3: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.psscan.PsScan

Investigators then analyzed the results to find suspicious processes. What was found almost immediately, was the keylogger-local.exe, which raised concerns just from the name alone. By extension, the parent process was immediately questioned as well.

PID	Image Name	Suspicious Indicator	Parent PID
3064	chrome.exe	Parent of cmd.exe	1576
488	cmd.exe	Parent of keylogger	3064 (chrome.exe)
2480	keylogger-local.exe	Malicious name – keylogger	488 (cmd.exe)
288	vmtoolsd.exe	Created duplicate instance	740
388	cmd.exe	Duplicate instance of 488	288 (vmtoolsd.exe)

Table 1: Suspicious Processes Identified

The keylogger-local.exe (PID 2308) appeared in the lists of pslist as well as psscan. It was also created immediately after its parent process was created. This process was spawned by cmd.exe (PID 488), which started and exited at the same time. There was also a duplicate of cmd.exe (PID 388) which was spawned by vmtoolsd.exe (288).

In order to further develop an understanding of the process intentions, investigators examined the contents of the command line history, using the windows.cmdline.CmdLine plugin (Figure 4).

windows.cmdline.CmdLine

Analysis of the DLL lists loaded to the suspected keylogger, and document was then completed to further develop an understanding of the investigation (Figure 5 & Figure 6).


```
1 Volatility 3 Framework 2.20.0
2
3 PID      Process Base   Size   Name      Path      LoadTime      File output
4
5 2308      keylogger-local 0x400000 0x6000 keylogger-local.exe C:\keylogger-local.exe N/A Disabled
6 2308      keylogger-local 0x7c900000 0xb2000 ntdll.dll C:\WINDOWS\system32\ntdll.dll N/A Disabled
7 2308      keylogger-local 0x7c800000 0xf6000 kernel32.dll C:\WINDOWS\system32\kernel32.dll N/A Disabled
8 2308      keylogger-local 0x77c10000 0x58000 msvcrt.dll C:\WINDOWS\system32\msvcrt.dll N/A Disabled
9 2308      keylogger-local 0x7e410000 0x91000 USER32.dll C:\WINDOWS\system32\USER32.dll N/A Disabled
10 2308      keylogger-local 0x77f10000 0x49000 GDI32.dll C:\WINDOWS\system32\GDI32.dll N/A Disabled
11 2308      keylogger-local 0x76390000 0x1d000 IMM32.DLL C:\WINDOWS\system32\IMM32.DLL N/A Disabled
12 2308      keylogger-local 0x77dd0000 0x9b000 ADVAPI32.dll C:\WINDOWS\system32\ADVAPI32.dll N/A Disabled
13 2308      keylogger-local 0x77e70000 0x93000 RPCRT4.dll C:\WINDOWS\system32\RPCRT4.dll N/A Disabled
14 2308      keylogger-local 0x77fe0000 0x11000 Secur32.dll C:\WINDOWS\system32\Secur32.dll N/A Disabled
15 2308      keylogger-local 0x74720000 0x4c000 MSCTF.dll C:\WINDOWS\system32\MSCTF.dll N/A Disabled
```

Figure 5: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.dllobject.DllList --pid 2308

```
1 Volatility 3 Framework 2.20.0
2
3 PID      Process Base   Size   Name      Path      LoadTime      File output
4
5 1456      WINWORD.EXE 0x30000000 0x66000 WINWORD.EXE C:\PROGRA-1\MICROS-2\Office12\WINWORD.EXE N/A Disabled
6 1456      WINWORD.EXE 0x7c900000 0xb2000 ntdll.dll C:\WINDOWS\system32\ntdll.dll N/A Disabled
7 1456      WINWORD.EXE 0x7c800000 0xf6000 kernel32.dll C:\WINDOWS\system32\kernel32.dll N/A Disabled
8 1456      WINWORD.EXE 0x77dd0000 0x9b000 ADVAPI32.dll C:\WINDOWS\system32\ADVAPI32.dll N/A Disabled
9 1456      WINWORD.EXE 0x77e70000 0x93000 RPCRT4.dll C:\WINDOWS\system32\RPCRT4.dll N/A Disabled
10 1456      WINWORD.EXE 0x77fe0000 0x11000 Secur32.dll C:\WINDOWS\system32\Secur32.dll N/A Disabled
11 1456      WINWORD.EXE 0x78130000 0x9b000 MSVCRT80.dll C:\WINDOWS\WinSxS\x86_Microsoft.VC80.CRT_1fc8b3b9a1e18e3b_8.0.50727.6195_x-ww_44262b86\MSVCRT80.dll N/A Disabled
12 1456      WINWORD.EXE 0x77c10000 0x58000 msvcrt.dll C:\WINDOWS\system32\msvcrt.dll N/A Disabled
13 1456      WINWORD.EXE 0x31240000 0x1146000 wwlib.dll C:\PROGRA-1\MICROS-2\Office12\wwlib.dll N/A Disabled
14 1456      WINWORD.EXE 0x77f10000 0x49000 GDI32.dll C:\WINDOWS\system32\GDI32.dll N/A Disabled
15 1456      WINWORD.EXE 0x7e410000 0x91000 USER32.dll C:\WINDOWS\system32\USER32.dll N/A Disabled
16 1456      WINWORD.EXE 0x774e0000 0x13e000 ole32.dll C:\WINDOWS\system32\ole32.dll N/A Disabled
17 1456      WINWORD.EXE 0x3a9d0000 0xf68000 oart.dll C:\PROGRA-1\MICROS-2\Office12\oart.dll N/A Disabled
18 1456      WINWORD.EXE 0x76390000 0x1d000 IMM32.DLL C:\WINDOWS\system32\IMM32.DLL N/A Disabled
19 1456      WINWORD.EXE 0x326e0000 0x1087000 mso.dll C:\Program Files\Common Files\Microsoft Shared\Office12\mso.dll N/A Disabled
20 1456      WINWORD.EXE 0x3fde0000 0x441000 msi.dll C:\WINDOWS\system32\msi.dll N/A Disabled
21 1456      WINWORD.EXE 0x33d00000 0xd7000 wwintl.dll C:\PROGRA-1\MICROS-2\Office12\1033\wwintl.dll N/A Disabled
22 1456      WINWORD.EXE 0x773d0000 0x103000 Comctl32.dll C:\WINDOWS\WinSxS\x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.2600.6028_x-ww_61e65202\Comctl32.dll N/A Disabled
23 1456      WINWORD.EXE 0x77f60000 0x76000 SHLWAPI.dll C:\WINDOWS\system32\SHLWAPI.dll N/A Disabled
24 1456      WINWORD.EXE 0x74720000 0x4c000 MSCTF.dll C:\WINDOWS\system32\MSCTF.dll N/A Disabled
25 1456      WINWORD.EXE 0x755c0000 0x2e000 mscftime.ime C:\WINDOWS\system32\mscftime.ime N/A Disabled
26 1456      WINWORD.EXE 0xd40000 0x655000 MSORES.DLL C:\Program Files\Common Files\Microsoft Shared\OFFICE12\MSORES.DLL N/A Disabled
27 1456      WINWORD.EXE 0x6bdc0000 0xba000 MSPTLS.DLL C:\Program Files\Common Files\Microsoft Shared\OFFICE12\MSPTLS.DLL N/A Disabled
28 1456      WINWORD.EXE 0x7c9c0000 0x817000 SHELL32.DLL C:\WINDOWS\system32\SHELL32.DLL N/A Disabled
29 1456      WINWORD.EXE 0x5d090000 0x9a000 comctl32.dll C:\WINDOWS\system32\comctl32.dll N/A Disabled
30 1456      WINWORD.EXE 0x1c80000 0x9dd000 MSOINTL.DLL C:\Program Files\Common Files\Microsoft Shared\Office12\1033\MSOINTL.DLL N/A Disabled
31 1456      WINWORD.EXE 0x79000000 0x4a000 mscoree.dll C:\WINDOWS\system32\mscoree.dll N/A Disabled
```

Figure 6: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.dllobject.DllList --pid 1456

Investigators then attempted to locate credible information from the results of utilizing the windows.malfind plugin (Figure 7). However, after analyzing the information provided, there was not any clear and definitive proof or evidence that furthered the investigation.

It was determined that utilizing the windows.vadinfo plugin would help investigators come to a concrete conclusion as to whether or not malware is present (Figure 8). After running the plugins, it was determined that there was an inconsistent and unusual file path associated with the keylogger.exe file.

[illegible]

2308

The same process was then completed for the WINWORD.exe file (Figure 9). Utilizing the windows.vadinfo plugin, investigators were able to utilize the information provided to provide direction of the investigation.

Volatility 3 Framework 2.10.0

PID	Process	Offset	Start VPM	End VPM	Tag	Protection	CommitCharge	PrivateMemory	Parent	File	File output
4											
5	1456	WINWORD.EXE	0x890ee958	0x51c0000	0x51c1fff	Vad	PAGE_READONLY	0	0	0x0	Disabled
6	1456	WINWORD.EXE	0x890ee1a8	0x4f20000	0x4f2ffff	VadS	PAGE_READWRITE	16	1	0x890ee958	N/A
7	1456	WINWORD.EXE	0x890efc9	0x4a20000	0x4a5ffff	VadS	PAGE_READWRITE	5	1	0x890ee1a8	N/A
8	1456	WINWORD.EXE	0x890ef38	0x4300000	0x437ffff	VadS	PAGE_READWRITE	8	1	0x890efc9	N/A
9	1456	WINWORD.EXE	0x890efc48	0x3570000	0x357ffff	VadS	PAGE_READWRITE	16	1	0x890ef38	N/A
10	1456	WINWORD.EXE	0x8a2c47a8	0x3510000	0x351ffff	VadS	PAGE_READWRITE	16	1	0x890efc48	N/A
11	1456	WINWORD.EXE	0x8909908	0x3190000	0x3198fff	VadS	PAGE_READWRITE	9	1	0x8a2c47a8	N/A
12	1456	WINWORD.EXE	0x8a202038	0x2f10000	0x2f1ffff	VadS	PAGE_READWRITE	16	1	0x8909908	N/A
13	1456	WINWORD.EXE	0x89f0c1b0	0x2a10000	0x2a3ffff	VadS	PAGE_READWRITE	16	1	0x8a202038	N/A
14	1456	WINWORD.EXE	0x8a3f7110	0x2a00000	0x2a2ffff	Vad	PAGE_EXECUTE_WRITECOPY	0	0	0x89f0c1b0	\\WINDOWS\system32\xpas2res.dll Disabled
15	1456	WINWORD.EXE	0x8908a808	0x1c00000	0x1c0cfff	VadL	PAGE_EXECUTE_WRITECOPY	0	0	0x8a3f7110	\\Program Files\Common Files\Microsoft Shared\OFFICE12\MSOINTL.DLL Disabled
16	1456	WINWORD.EXE	0x8a31f6c0	0x1550000	0x155ffff	VadS	PAGE_READWRITE	16	1	0x8908a808	N/A
17	1456	WINWORD.EXE	0x89082208	0x1440000	0x144ffff	VadL	PAGE_READWRITE	16	1	0x8a31f6c0	N/A
18	1456	WINWORD.EXE	0x89092508	0x13c0000	0x13c8fff	VadL	PAGE_READWRITE	1	1	0x89082208	N/A
19	1456	WINWORD.EXE	0x890ef678	0xc0000	0xc1ffff	VadS	PAGE_READWRITE	16	1	0x89092508	N/A
20	1456	WINWORD.EXE	0x89091528	0x990000	0x99ffff	VadL	PAGE_WACCESS	1	1	0x890ef678	N/A
21	1456	WINWORD.EXE	0x8a3dd0a8	0x920000	0x92ffff	VadS	PAGE_READWRITE	16	1	0x89091528	N/A
22	1456	WINWORD.EXE	0x8908a008	0x80000	0x808fff	VadL	PAGE_READWRITE	1	1	0x8a3dd0a8	N/A
23	1456	WINWORD.EXE	0x89f01550	0x3550000	0x352fff	Vad	PAGE_READONLY	0	0	0x8908a008	\\WINDOWS\system32\ctype.nls Disabled
24	1456	WINWORD.EXE	0x8908b7e8	0x250000	0x25ffff	VadS	PAGE_READWRITE	0	1	0x89f01550	N/A
25	1456	WINWORD.EXE	0x89090808	0x20000	0x12ffff	VadL	PAGE_READWRITE	49	1	0x8908b7e8	N/A
26	1456	WINWORD.EXE	0x89f07040	0x18000	0x11ffff	VadS	PAGE_READWRITE	2	1	0x89090808	N/A
27	1456	WINWORD.EXE	0x89f0b550	0x20000	0x20ffff	VadL	PAGE_READWRITE	1	1	0x89f07040	N/A
28	1456	WINWORD.EXE	0x8a235208	0x140000	0x140fff	Vad	PAGE_READONLY	0	0	0x89f0b550	N/A
29	1456	WINWORD.EXE	0x8a2c4008	0x130000	0x132fff	Vad	PAGE_READONLY	0	0	0x8a235208	N/A
30	1456	WINWORD.EXE	0x8a30b418	0x150000	0x154fff	VadL	PAGE_READWRITE	139	1	0x8a2c4008	N/A
31	1456	WINWORD.EXE	0x89fa2a28	0x20000	0x208fff	Vad	PAGE_READONLY	0	0	0x8a30b418	\\WINDOWS\system32\sortkey.nls Disabled
32	1456	WINWORD.EXE	0x89092008	0x270000	0x265fff	Vad	PAGE_READONLY	0	0	0x89fa2a28	\\WINDOWS\system32\unicode.nls Disabled
33	1456	WINWORD.EXE	0x8a30b418	0x260000	0x264fff	Vad	PAGE_READWRITE	0	0	0x89092008	N/A
34	1456	WINWORD.EXE	0x8908b1a8	0x290000	0x2d0fff	Vad	PAGE_READONLY	0	0	0x89092008	\\WINDOWS\system32\locale.nls Disabled
35	1456	WINWORD.EXE	0x89060708	0x340000	0x34ffff	VadS	PAGE_READWRITE	9	1	0x89fa2a28	N/A
36	1456	WINWORD.EXE	0x89092008	0x320000	0x323fff	Vad	PAGE_READONLY	0	0	0x89060708	\\WINDOWS\system32\sorttbls.nls Disabled
37	1456	WINWORD.EXE	0x89f02008	0x390000	0x457fff	Vad	PAGE_EXECUTE_READ	0	0	0x89092008	N/A
38	1456	WINWORD.EXE	0x89f07720	0x370000	0x373fff	Vad	PAGE_READONLY	0	0	0x89f02008	N/A
39	1456	WINWORD.EXE	0x890a6078	0x360000	0x36ffff	VadS	PAGE_READWRITE	16	1	0x89f07720	N/A
40	1456	WINWORD.EXE	0x890e1550	0x380000	0x383fff	Vad	PAGE_READONLY	0	0	0x890a6078	N/A
41	1456	WINWORD.EXE	0x8901dc0	0x580000	0x87ffff	Vad	PAGE_EXECUTE_READ	0	0	0x890e1550	N/A
42	1456	WINWORD.EXE	0x8908140	0x480000	0x562fff	Vad	PAGE_READONLY	0	0	0x8901dc0	N/A
43	1456	WINWORD.EXE	0x89f01550	0x570000	0x570fff	VadL	PAGE_READWRITE	1	1	0x8908140	N/A
44	1456	WINWORD.EXE	0x8a056e08	0x80000	0x80dfff	Vad	PAGE_READWRITE	0	0	0x89f01550	N/A
45	1456	WINWORD.EXE	0x8909e720	0x80000	0x8cfff	VadS	PAGE_READWRITE	43	1	0x8a056e08	N/A
46	1456	WINWORD.EXE	0x8a30b508	0x890000	0x89ffff	VadS	PAGE_READWRITE	16	1	0x8909e720	N/A
47	1456	WINWORD.EXE	0x8a202038	0x90000	0x8dfff	Vad	PAGE_READONLY	0	0	0x8a30b508	N/A
48	1456	WINWORD.EXE	0x89f5c410	0x90000	0x917fff	VadS	PAGE_READWRITE	24	1	0x8a202038	N/A
49	1456	WINWORD.EXE	0x8a3f7110	0x8f0000	0x8f3fff	VadS	PAGE_READWRITE	4	1	0x89f5c410	N/A
50	1456	WINWORD.EXE	0x89f0b700	0x90000	0x90ffff	VadS	PAGE_READWRITE	16	1	0x8a3f7110	N/A
51	1456	WINWORD.EXE	0x8a325208	0x940000	0x94ffff	VadS	PAGE_WACCESS	2	1	0x89f0b700	N/A
52	1456	WINWORD.EXE	0x89092508	0x930000	0x930fff	VadS	PAGE_READWRITE	1	1	0x8a325208	N/A
53	1456	WINWORD.EXE	0x89092008	0x950000	0x95ffff	VadS	PAGE_READWRITE	16	1	0x89092508	N/A
54	1456	WINWORD.EXE	0x89092508	0x980000	0x98ffff	VadL	PAGE_READWRITE	16	1	0x89092008	N/A
55	1456	WINWORD.EXE	0x89060008	0x970000	0x97ffff	VadS	PAGE_READWRITE	16	1	0x89092508	N/A
56	1456	WINWORD.EXE	0x890a7408	0x90000	0x9dffff	VadL	PAGE_READWRITE	16	1	0x89060008	N/A
57	1456	WINWORD.EXE	0x89097770	0x90000	0x90ffff	VadS	PAGE_READWRITE	16	1	0x890a7408	N/A

Figure 9: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.vadinfo --pid

1456

Identification of potentially suspicious memory regions, while also determining off-putting file paths, investigators used the windows.pedump plugin to determine whether or not the file can be considered malware and confirm initial suspicions (Figure 10 & Figure 11).

PID	Process	File output
2308	keylogger-local	PE.0x8969d638.2308.0x400000.dmp

Figure 10: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.pedump --pid

2308 --base 0x400000

PID	Process	File	output
1456	WINWORD.EXE	PE.0x8a36bda0.1456.0x30000000.dmp	

Figure 11: `python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.pedump --pid 1456 --base 0x30000000`

Using the windows.vadinfo plugin in conjunction with the windows.pedump plugin, the results helped confirm that keylogger-local.exe had executable memory not linked to any DLL and suspicious custom regions.

From this point, investigators then attempted to utilize the dumps provided to determine whether or not malware was present, using VirusTotal. Through the analysis provided by this tool, it was determined that malware was present in multiple locations (Figure 12 & Figure 13).

20/72 security vendors flagged this file as malicious

0cf07e5944bc3a719d605acaa8698aa9abadc1a90df3b7fd4ef5041016a627f8
PE.0x8969d638.2308.0x400000.dmp

Size: 24.00 KB | Last Analysis Date: 25 days ago

peexe checks-user-input idle long-sleeps

Community Score: 20 / 72

Popular threat label: **trojan.keylogger** | Threat categories: trojan | Family labels: keylogger

Security vendors' analysis

Vendor	Detection
Arctic Wolf	Unsafe
AVG	Win32:Malware-gen
CrowdStrike Falcon	Win/malicious_confidence_90% (D)
Avast	Win32:Malware-gen
Avira (no cloud)	TR/Crypt.XPACK.Gen
Cynet	Malicious (score: 100)

Figure 12: VirusTotal Scan Keylogger

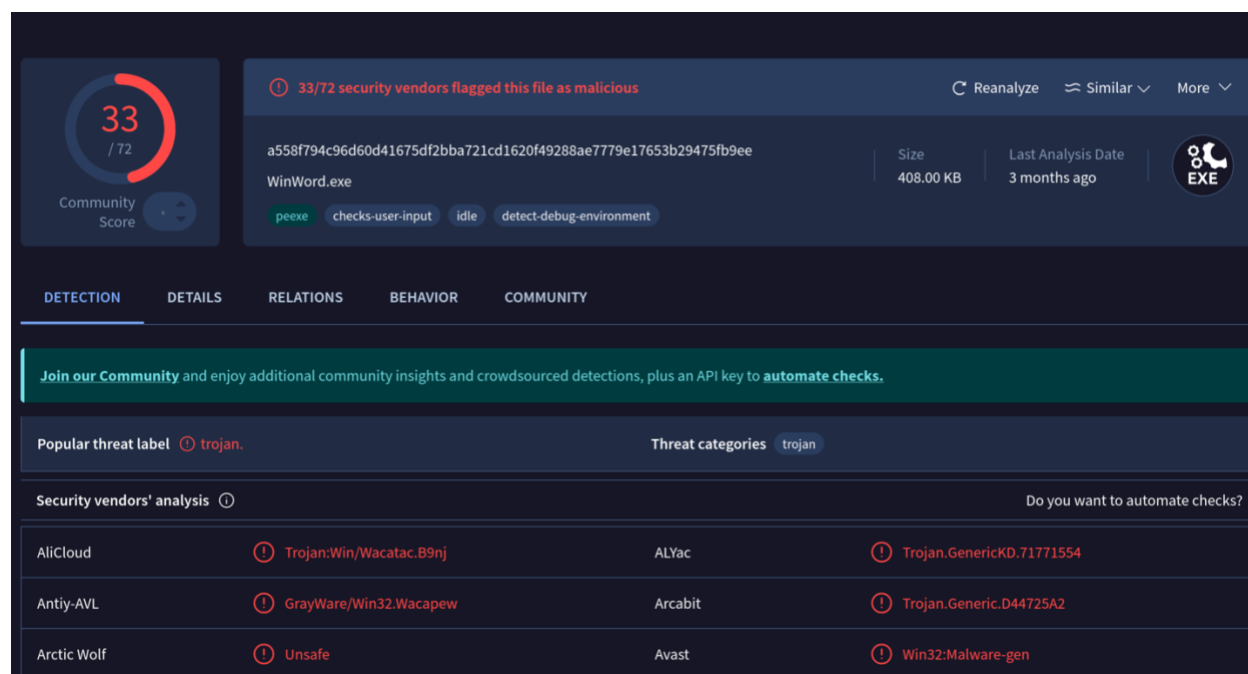


Figure 13: VirusTotal Scan WINWORD

The WINWORD.exe file seem to likely be the initial loader or the injected payload from WINWORD.exe. It appears packed or obfuscated and may have dropped or unpacked keylogger-local.exe to disk or memory. The keylogger-local.exe file has now been confirmed to be a form of keylogger malware. According to the analysis, that most likely employed some form of hooking techniques. Obviously, logging keystrokes and potentially exfiltrating this data would most likely be its primary purpose.

Investigators then attempted to dig deeper into network analysis. Investigators initially began attempting to utilize plugins such as windows.netscan and windows.netstat in order to divulge many of the possible connections that may have been established on this system. However, many issues were encountered throughout this process using Volatility 3. As an

alternative, Volatility 2 was used in order to attempt to find this information, which resulted in proper functionality. Beginning with the connections plugin, investigators were able to list all active TCP connections belonging to the system (Figure 14).

```
Time Stamp: Wed Apr 30 11:33:28 2025
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\volatility.exe" --plugins=
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\profiles" connections --filename=
"C:\Users\chris\Desktop\maleficent.vmem" --profile=WinXPSP2x86 --kdbg=0x80545ce0

Please wait, this may take a few minutes.

Time Stamp: Wed Apr 30 11:33:42 2025
Volatility Foundation Volatility Framework 2.6.1
Offset(V) Local Address Remote Address Pid
-----
0x89977600 127.0.0.1:1521 127.0.0.1:1047 132
0x8a296008 127.0.0.1:1047 127.0.0.1:1521 224
0x8988c8d8 127.0.0.1:5152 127.0.0.1:1309 1768
0x8a43e2b8 127.0.0.1:1309 127.0.0.1:5152 3852
0x8a23fb40 127.0.0.1:1248 127.0.0.1:22350 1140

Time Stamp: Wed Apr 30 11:33:42 2025

***** End of command output *****
```

Figure 14: Vol2 connections

What was found was five different Local Address, Remote Address, and PID of processes with TCP connections. This provided investigators a baseline of what to expect for future network activity. Following the results of the connections plugin, the connscan plugin was used (Figure 15).

```

Time Stamp: Wed Apr 30 12:37:05 2025
"C:\Users\ghis\Downloads\Volatility Workbench-v2.1\volatility.exe" --plugin=
"C:\Users\ghis\Downloads\Volatility Workbench-v2.1\profiles" --plugin=connscan --filename=
"C:\Users\ghis\Desktop\malware\vmem" --profile=WinXPSP2x86 --kdbg=0x80545ce0

Please wait, this may take a few minutes.

Time Stamp: Wed Apr 30 12:37:21 2025
Volatility Foundation Volatility Framework 2.6.1
Offset(P) Local Address Remote Address Pid
-----
0x095ff238 172.16.156.246:1409 72.21.91.121:80 3852
0x095ff438 172.16.156.246:1434 23.212.53.165:443 3852
0x09600008 172.16.156.246:1419 192.0.78.23:443 3852
0x09600008 172.16.156.246:1419 192.0.78.23:443 3852
0x096005b8 172.16.156.246:1422 108.174.2.129:80 3852
0x09600d08 172.16.156.246:1430 23.212.53.165:443 3852
0x0961ee68 172.16.156.246:1428 23.212.53.165:443 3852
0x09620b68 172.16.156.246:1427 23.212.53.165:443 3852
0x09620e68 172.16.156.246:1397 72.21.91.101:80 3852
0x096216b0 3.0.103.2:26996 93.184.216.146:22369 2304732552
0x096263b8 172.16.156.246:1410 192.0.82.250:80 3852
0x096298b8 172.16.156.246:1406 72.21.91.111:80 3852
0x0962b8b8 172.16.156.246:1408 173.194.115.0:80 3852
0x0962ce68 172.16.156.246:1429 23.212.53.222:443 3852
0x0962eb58 172.16.156.246:1411 192.0.76.3:80 3852
0x0963b250 172.16.156.246:1416 23.212.53.228:80 3852
0x09647d68 172.16.156.246:1383 23.235.44.175:443 3852
0x09649468 172.16.156.246:1433 23.212.53.222:443 3852
0x09649b60 224.219.97.137:0 248.106.134.137:512 2
0x0964ce68 172.16.156.246:1420 72.21.91.121:80 3852
0x096553d8 172.16.156.246:1338 173.194.115.20:80 3852
0x09655578 172.16.156.246:1330 12.0.1.0:443 2317026792
0x09659470 172.16.156.246:1301 23.212.53.204:80 2360
0x09659a40 172.16.156.246:1300 23.212.53.188:80 2360
0x0965c530 172.16.156.246:1326 173.194.115.0:443 3852
0x09661e68 172.16.156.246:1414 72.21.91.111:80 3852
0x09667c08 172.16.156.246:1378 74.217.81.215:80 3852
0x0966a008 172.16.156.246:1359 23.212.53.196:80 3852
0x09676008 172.16.156.246:1312 173.194.115.31:443 3852
0x0967dbb8 172.16.156.246:1327 173.194.115.0:443 3852
0x096802a0 172.16.156.246:1337 173.194.115.0:443 3852
0x09682008 172.16.156.246:1404 192.254.231.187:80 3852
0x0969f008 172.16.156.246:1322 173.194.115.5:443 3852
0x096a0850 160.221.42.138:11146 112.63.43.138:26687 2305452144
0x096a6c00 172.16.156.246:1280 173.194.115.2:80 2360
0x09830b90 172.16.156.246:1421 31.13.66.128:443 3852
0x09832e68 172.16.156.246:1405 173.194.115.31:80 3852
0x09834b90 172.16.156.246:1415 192.0.76.3:80 3852
0x09836858 172.16.156.246:1426 23.212.53.165:443 3852
0x09838158 172.16.156.246:1385 50.31.164.166:443 3852
0x0983dac0 19.0.0.0:25225 48.114.250.137:46 16777235
0x098417e8 172.16.156.246:1413 172.233.38.82:80 3852
0x09843468 172.16.156.246:1432 23.212.53.165:443 3852
0x09848250 172.16.156.246:1417 172.233.48.156:443 3852
0x09857498 172.16.156.246:1379 74.217.81.213:443 3852
0x09864888 172.16.156.246:1373 108.168.240.194:80 3852
0x09865858 172.16.156.246:1366 173.194.115.25:80 3852
0x09867468 172.16.156.246:1399 192.254.231.187:80 3852
0x0986e008 172.16.156.246:1353 173.194.115.25:80 3852
0x09872e68 172.16.156.246:1331 173.194.115.7:443 3852
0x0987de68 172.16.156.246:1352 173.194.115.26:80 3852
0x0987ee68 172.16.156.246:1333 173.194.115.7:443 3852
0x09881868 172.16.156.246:1336 173.194.115.5:443 3852
0x09883e68 172.16.156.246:1349 173.194.115.30:80 3852

```

Figure 15: Vol2 connscan

The connscan plugin was able to successfully carve network connection artifacts from memory (both live and closed connections), which provided investigators with a much more intensive understanding of network-related evidence. The sockets and sockscan plugins, which was the next plugin that was used further helped develop an understanding of network-based evidence on the system being investigated (Figure 16 & Figure 17).


```

"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\volatility.exe" --plugin=
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\profiles\mcafee-slimmer"
"C:\Users\chris\Desktop\molefense.mmon" --profile=WnXPSP2x86 --klib=0x80545c0d
Please wait, this may take a few minutes.

Time Stamp: Wed Apr 30 12:37:39 2025
Volatility Foundation Volatility Framework 2.6.1
Offset(V) PID Port Proto Protocol Address Create Time
0x896356d0 3852 1398 6 TCP 0.0.0.0 2014-10-29 17:18:58 UTC+0000
0x895d588 3852 1433 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89844e98 3852 1402 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x8992548 3852 1309 6 TCP 0.0.0.0 2014-10-29 17:18:25 UTC+0000
0x8999e98 4 0 47 GRE 0.0.0.0 2014-10-27 20:01:35 UTC+0000
0x8a194008 1684 2250 17 UDP 0.0.0.0 2014-10-27 20:01:26 UTC+0000
0x89efb98 200 22 6 TCP 0.0.0.0 2014-10-27 20:01:28 UTC+0000
0x896eb30 3852 1410 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x8a32a78 752 509 17 UDP 0.0.0.0 2014-10-27 20:01:28 UTC+0000
0x89628948 3852 1418 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x89946910 1820 3306 6 TCP 0.0.0.0 2014-10-27 20:01:29 UTC+0000
0x896281e8 3852 1422 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8a722610 4 445 6 TCP 0.0.0.0 2014-10-27 20:01:18 UTC+0000
0x8983d880 3852 1426 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x8a266e98 1040 135 6 TCP 0.0.0.0 2014-10-27 20:01:21 UTC+0000
0x89628008 3852 1399 6 TCP 0.0.0.0 2014-10-29 17:18:59 UTC+0000
0x895f5e8 3852 1434 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89827a60 3852 1403 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89693e98 1140 1248 6 TCP 0.0.0.0 2014-10-29 17:01:53 UTC+0000
0x898844e8 3852 1305 17 UDP 127.0.0.1 2014-10-29 17:18:25 UTC+0000
0x8962c8 1684 2250 6 TCP 0.0.0.0 2014-10-27 20:01:26 UTC+0000
0x8983e98 3852 1411 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x89977e98 1508 1040 17 UDP 0.0.0.0 2014-10-27 20:02:16 UTC+0000
0x8962670 3852 1415 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8a265008 1180 1437 17 UDP 127.0.0.1 2014-10-29 17:20:02 UTC+0000
0x8a35d860 200 1025 6 TCP 127.0.0.1 2014-10-27 20:01:28 UTC+0000
0x8982eb28 3852 1419 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8981c008 1180 123 17 UDP 127.0.0.1 2014-10-29 17:20:02 UTC+0000
0x8a3a358 224 1047 6 TCP 0.0.0.0 2014-10-27 20:02:33 UTC+0000
0x8a3647e0 752 0 255 Reserved 0.0.0.0 2014-10-27 20:01:28 UTC+0000
0x8984b340 3852 1427 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89627830 3852 1400 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89634e8 3852 1431 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x8983e90 3852 1404 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x8a403498 224 1028 6 TCP 0.0.0.0 2014-10-27 20:01:28 UTC+0000
0x895f758 3852 1408 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x89614500 3852 1412 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x89883368 4 1036 6 TCP 0.0.0.0 2014-10-27 20:01:35 UTC+0000
0x89841e08 3852 1416 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8a34438 132 1026 6 TCP 127.0.0.1 2014-10-27 20:01:28 UTC+0000
0x8996008 3852 1424 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8998840 1272 1034 6 TCP 127.0.0.1 2014-10-27 20:01:34 UTC+0000
0x8986910 3852 1428 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x898d158 132 1521 6 TCP 0.0.0.0 2014-10-27 20:01:28 UTC+0000
0x8962930 3852 1432 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89836270 3852 1401 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89871688 3852 1405 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x8a29008 1260 1900 17 UDP 127.0.0.1 2014-10-29 17:20:02 UTC+0000
0x8917e98 752 4500 17 UDP 0.0.0.0 2014-10-27 20:01:28 UTC+0000
0x8968250 3852 1417 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x896eb718 1232 1227 6 TCP 0.0.0.0 2014-10-29 16:54:42 UTC+0000
0x8992b78 1768 5152 6 TCP 127.0.0.1 2014-10-27 20:01:29 UTC+0000
0x8a72780 4 445 17 UDP 0.0.0.0 2014-10-27 20:01:18 UTC+0000

```

Figure 16: Vol2 sockets

```

Time Stamp: Wed Apr 30 12:38:03 2025
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\volatility.exe" --plugin=
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\profiles\mcafee-slimmer"
"C:\Users\chris\Desktop\molefense.mmon" --profile=WnXPSP2x86 --klib=0x80545c0d
Please wait, this may take a few minutes.

Time Stamp: Wed Apr 30 12:38:18 2025
Volatility Foundation Volatility Framework 2.6.1
Offset(P) PID Port Proto Protocol Address Create Time
0x895f6758 3852 1408 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x895f688 3852 1433 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x895f688 3852 1433 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x895f758 3852 1434 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89614500 3852 1412 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x8961478 3852 1406 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x89624e98 3852 1429 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89627330 3852 1400 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89628008 3852 1399 6 TCP 0.0.0.0 2014-10-29 17:18:59 UTC+0000
0x896281e8 3852 1422 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x89628948 3852 1418 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8962a70 3852 1409 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x896356d0 3852 1398 6 TCP 0.0.0.0 2014-10-29 17:18:58 UTC+0000
0x89635d8 3852 1431 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89632930 3852 1432 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89632a08 3852 1348 6 TCP 0.0.0.0 2014-10-29 17:18:43 UTC+0000
0x89635870 3852 1357 6 TCP 0.0.0.0 2014-10-29 17:18:44 UTC+0000
0x89657908 3852 1315 6 TCP 0.0.0.0 2014-10-29 17:18:34 UTC+0000
0x8965a778 593920 52 0 HOPORT 0.0.0.0 2014-10-29 17:18:45 UTC+0000
0x8965884 3852 1230 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89658a38 3852 1430 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89658d98 3852 1327 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89660a08 3852 1368 6 TCP 0.0.0.0 2014-10-29 17:18:44 UTC+0000
0x89660808 3852 1360 6 TCP 0.0.0.0 2014-10-29 17:18:44 UTC+0000
0x8966a218 221_92 28818 33174 144.192.65.138 2014-10-29 17:18:44 UTC+0000
0x89676c14 3852 1311 6 TCP 0.0.0.0 2014-10-29 17:18:26 UTC+0000
0x89680378 3852 1337 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89682900 3852 1432 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89682e50 3852 1417 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x89685880 3852 1325 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89693e98 1140 1248 6 TCP 0.0.0.0 2014-10-29 17:01:53 UTC+0000
0x896a5714 1232 1227 6 TCP 0.0.0.0 2014-10-29 16:54:42 UTC+0000
0x896a6258 3852 1419 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x89692660 3852 1403 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89833e98 3852 1411 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x89836270 3852 1401 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89836e98 3852 1421 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x8983c008 1180 123 17 UDP 127.0.0.1 2014-10-29 17:20:02 UTC+0000
0x8983c098 3852 1383 6 TCP 0.0.0.0 2014-10-29 17:18:51 UTC+0000
0x89834880 3852 1426 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x8983c930 3852 1404 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89841e08 3852 1416 6 TCP 0.0.0.0 2014-10-29 17:19:02 UTC+0000
0x89844e98 3852 1402 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89846340 3852 1427 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89847220 3852 1379 6 TCP 0.0.0.0 2014-10-29 17:18:45 UTC+0000
0x89866e98 3852 1364 6 TCP 0.0.0.0 2014-10-29 17:18:44 UTC+0000
0x89866b30 3852 1410 6 TCP 0.0.0.0 2014-10-29 17:19:01 UTC+0000
0x89868110 3852 1428 6 TCP 0.0.0.0 2014-10-29 17:19:03 UTC+0000
0x89871488 3852 1405 6 TCP 0.0.0.0 2014-10-29 17:19:00 UTC+0000
0x89872208 3852 1331 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89872208 3852 1332 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89876008 3852 1333 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000
0x89882900 3852 1326 6 TCP 0.0.0.0 2014-10-29 17:18:40 UTC+0000

```

Figure 17: Vol 2 sockscan

The sockets and sockscan plugin provided a great understanding of what was happening on the system, by also providing a timeline of events based on the Create_Time data provided by the plugin.

Investigators finished the acquisition of network evidence by attempting to utilize the netscan plugin (Figure 18). However, unfortunately the profile used to enumerate prior network information did not allow for the plugin to operate as intended.

```
Time Stamp: Wed Apr 30 12:38:22 2025
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\volatility.exe" --plugins=
"C:\Users\chris\Downloads\VolatilityWorkbench-v2.1\profiles" netscan --filename=
"C:\Users\chris\Desktop\maleficent.vmem" --profile=WinXPSP2x86 --kdbg=0x80545ce0

Please wait, this may take a few minutes.

Time Stamp: Wed Apr 30 12:38:32 2025
Volatility Foundation Volatility Framework 2.6.1
ERROR : volatility.debug : This command does not support the profile WinXPSP2x86

Time Stamp: Wed Apr 30 12:38:32 2025

***** End of command output *****
```

Figure 18: Vol2 netscan

The attention of the investigation was then switched to focus on the potential to acquire information related to images that may be found on the system. An initial search was conducted using the windows.filescan plugin on the memory image was conducted to focus on select file types, such as the .jpg and .png extensions (Figure 19).

```
1 0x95f0b6a Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
2 0x95f0b6b Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
3 0x95f0b6c Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
4 0x95f0b6d Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
5 0x95f0b6e Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
6 0x95f0b6f Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
7 0x95f0b70 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
8 0x95f0b71 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
9 0x95f0b72 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
10 0x95f0b73 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
11 0x95f0b74 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
12 0x95f0b75 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
13 0x95f0b76 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
14 0x95f0b77 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
15 0x95f0b78 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
16 0x95f0b79 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
17 0x95f0b7a Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
18 0x95f0b7b Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
19 0x95f0b7c Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
20 0x95f0b7d Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
21 0x95f0b7e Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
22 0x95f0b7f Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
23 0x95f0b80 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
24 0x95f0b81 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
25 0x95f0b82 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
26 0x95f0b83 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
27 0x95f0b84 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
28 0x95f0b85 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
29 0x95f0b86 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
30 0x95f0b87 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
31 0x95f0b88 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
32 0x95f0b89 Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\13L88V78\782_0_004511.jpg
```

Figure 19: python3 vol.py -f /home/relik/Downloads/maleficent.vmem windows.filescan.FileScan | grep -Ei "\.jpe?g|.png|.gif|.bmp"

From this point, investigators were able to summarize the results into a text file containing only the offsets of each of the files (Figure 20).

```
0x95fbb68
0x95fbc68
0x95fc250
0x95fc5f0
0x95fcb58
0x95fcbf0
0x95fcc88
0x95fce58
0x95fd1b8
0x95fd250
0x95fd2f0
0x95fd420
0x95fd7b8
0x95fd850
0x95fd8e8
0x95fd988
0x95fda20
0x95dab8
0x95fdb58
0x95fdbf0
0x95fdc88
0x95fdd28
0x95fddc0
0x95fdf00
0x95fe920
0x95feb30
0x95fee58
0x9600a50
0x961d198
0x961db58
```

Figure 20: Contents of offsets.txt

With all information available, investigators created a simple python script to run and dump all files gathered based on the offsets. However, after encountering empty results, it was determined that the usage of Volatility 3 was once again inadequate to obtain the results (Figure 21).

```
(relik@kali)-[~/Downloads]
$ chmod +x extract_images_from_offsets.sh

(relik@kali)-[~/Downloads]
$ ./extract_images_from_offsets.sh
[*] Dumping from offset 0x95fbb68 ...
[*] Dumping from offset 0x95fbc68 ...
[*] Dumping from offset 0x95fc250 ...
[*] Dumping from offset 0x95fc5f0 ...
[*] Dumping from offset 0x95fcb58 ...
[*] Dumping from offset 0x95fcbf0 ...
[*] Dumping from offset 0x95fcc88 ...
[*] Dumping from offset 0x95fce58 ...
```

Figure 21: Failed Volatility Usage

Investigators then employed Volatility 2 once again, and repeated the aforementioned process, and creating a new script (Figure 22).

```
#!/bin/bash

VOLATILITY_PATH="/home/relik/volatility-2.6.1/vol.py"
MEMORY_IMAGE="/home/relik/Downloads/mafeicent.vmem"
PROFILE="WinXPS2x86"
OFFSETS_FILE="offsets.txt"
OUTPUT_DIR="dumped_images"

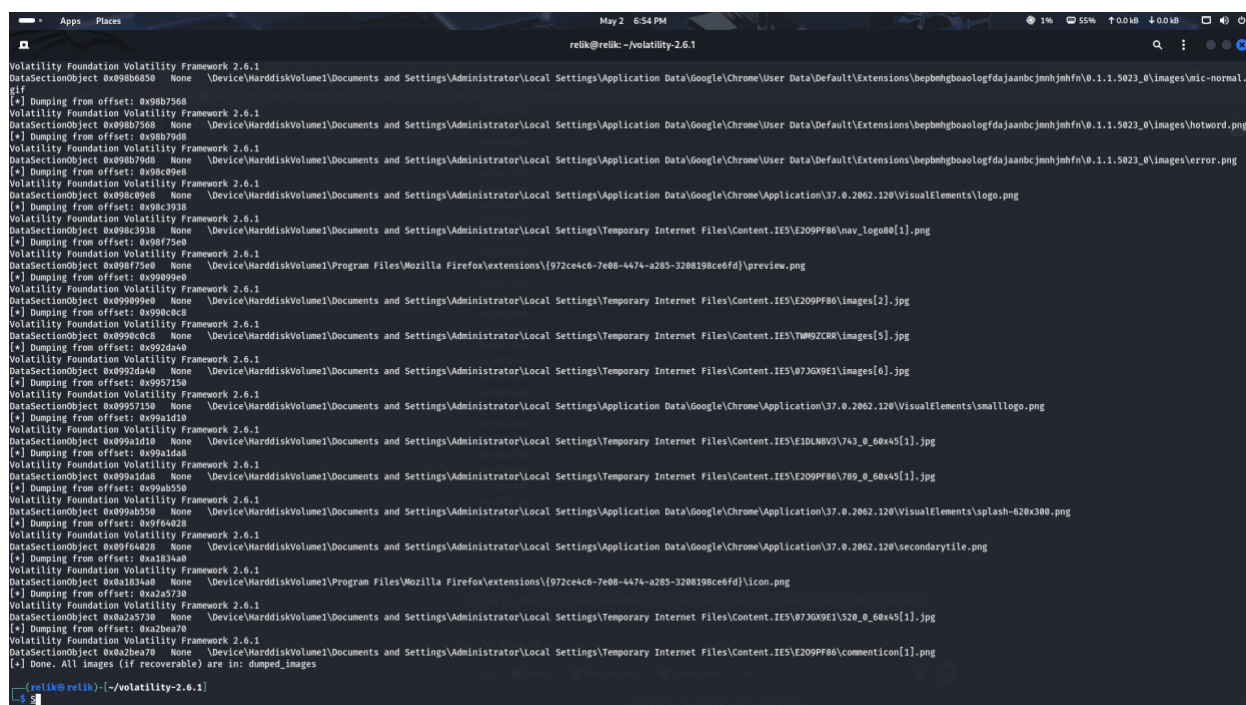
mkdir -p "$OUTPUT_DIR"

while read -r offset; do
    echo "[*] Dumping from offset: $offset"
    python2 "$VOLATILITY_PATH" -f "$MEMORY_IMAGE" --profile="$PROFILE" dumpfiles -Q "$offset" -D
"$OUTPUT_DIR"
done < "$OFFSETS_FILE"

echo "[+] Done. All images (if recoverable) are in: $OUTPUT_DIR"
```

Figure 22: Simple Python Script For Mass Dump of Images

From this point, investigators were able to determine that the commands would function properly, and employed the script to obtain a batch copy of all images on the device (Figure 23).



```
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x098b850 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\User Data\Default\Extensions\bepbmhgbaolofdjaanbcjnmhjhfn\0.1.1.5823_0\images\mic-normal.
gif
[*] Dumping from offset: 0x98b7568
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x098b7568 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\User Data\Default\Extensions\bepbmhgbaolofdjaanbcjnmhjhfn\0.1.1.5823_0\images\hotword.png
[*] Dumping from offset: 0x98b79d8
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x098b79d8 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\User Data\Default\Extensions\bepbmhgbaolofdjaanbcjnmhjhfn\0.1.1.5823_0\images\error.png
[*] Dumping from offset: 0x98c09e8
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x098c09e8 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\Application\37.0.2062.120\VisualElements\logo.png
[*] Dumping from offset: 0x98c3938
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x098c3938 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\E209PF86\nav_log00[1].png
[*] Dumping from offset: 0x98f75e0
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x098f75e0 None \Device\HarddiskVolume\Program Files\Mozilla Firefox\extensions\{972ce4c6-7e08-4474-a285-3208198ce6fd}\preview.png
[*] Dumping from offset: 0x99090e0
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x99090e0 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\TW9ZCRR\images[5].jpg
[*] Dumping from offset: 0x992d440
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x992d440 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\0730X9E1\images[6].jpg
[*] Dumping from offset: 0x9957150
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x9957150 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\Application\37.0.2062.120\VisualElements\smalllogo.png
[*] Dumping from offset: 0x99a1d10
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x99a1d10 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\EIDLNV3\743_0_60x45[1].jpg
[*] Dumping from offset: 0x99a1d88
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x99a1d88 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\E209PF86\789_0_60x45[1].jpg
[*] Dumping from offset: 0x99ab550
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x99ab550 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\Application\37.0.2062.120\VisualElements\splash-620x300.png
[*] Dumping from offset: 0x9ff64028
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x9ff64028 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Application Data\Google\Chrome\Application\37.0.2062.120\secondarytile.png
[*] Dumping from offset: 0xa1834a0
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0xa1834a0 None \Device\HarddiskVolume\Program Files\Mozilla Firefox\extensions\{972ce4c6-7e08-4474-a285-3208198ce6fd}\icon.png
[*] Dumping from offset: 0xa2a5730
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0xa2a5730 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\0730X9E1\S20_0_60x45[1].jpg
[*] Dumping from offset: 0xa2bea70
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0xa2bea70 None \Device\HarddiskVolume\Documents and Settings\Administrator\Local Settings\Temporary Internet Files\Content.IE5\E209PF86\commenticon[1].png
[*] Done. All images (if recoverable) are in: dumped_images

(relik@relik) ~/volatility-2.6.1
```

Figure 23: Successful Utilization of Volatility 2 Python Script

All images were properly dumped into a directory labeled 'dumped_images'. From here, investigators parsed through the images to locate evidence based on the leads provided in the instructions (Figure 24).

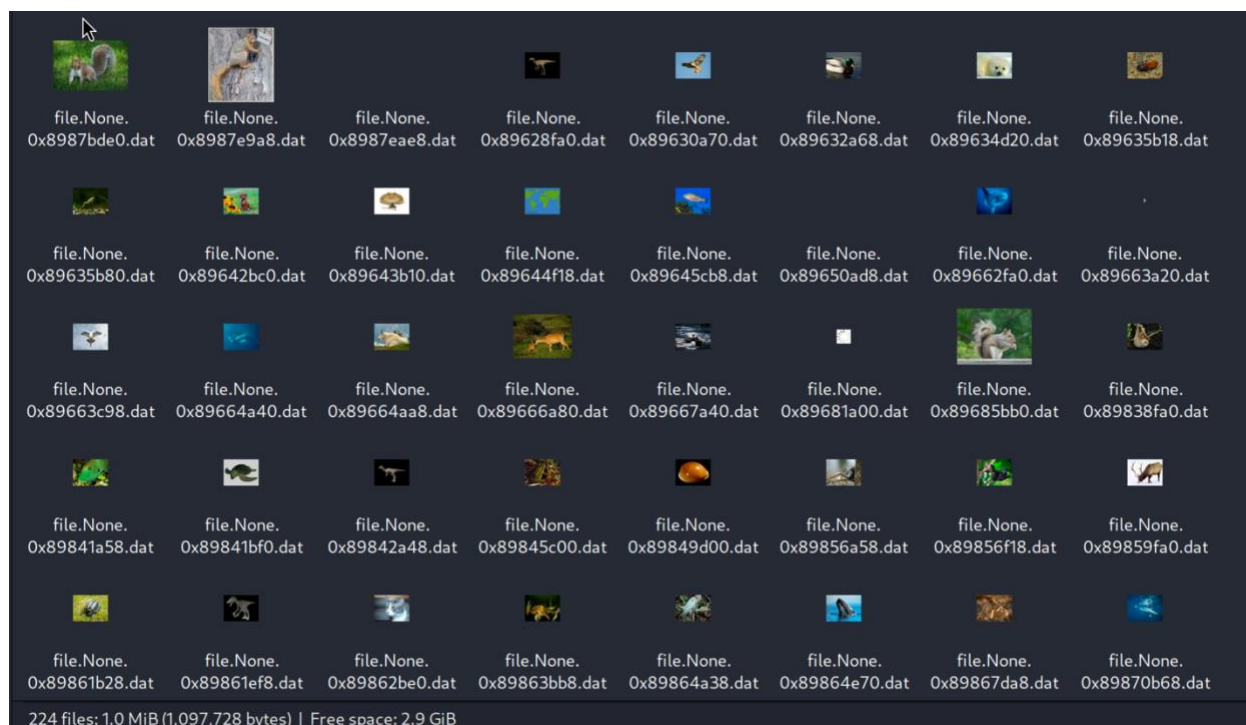


Figure 24: Section of Dumped Images Displaying Animals

Throughout the analysis of this image, it was also noted that a file was found after locating the WINWORD.exe file, and was also noted in the same parent directory of the directory containing the images. This file was labeled as 'Dear Mother.docx' (Figure 25).

Dear Mother,

This nice professor, Golden, recently loaned me one of his laptops so I can do some work while he takes a few days away at the beach to unwind. Finally, I can have a computer all to myself, to do whatever I want with it! He even changed his passwords (I watched him do it) and said that he would wipe the laptop when he returned so everything I do on it is completely private! It's awesome!

Talk to you later.

--Mary

Figure 25: Contents of Dear_Mother.docx

At this point, investigators were confident that all evidence related to the investigation was produced in a forensically sound manner.

Challenges & Responses

Throughout the acquisition process, investigators encountered issues running certain commands within Volatility3. This was due to reasons unknown initially, however after further analysis, it became clear that due to the file type, it was not possible to gather information related to network statistics and console history. These issues were remediated through the usage of Volatility 2, a precursor to Volatility 3 with the full capability of running all plugins not yet optimized for Volatility 3. This allowed investigators to continue the acquisition process without fear of missing out on potential evidence that could be used to further the investigation.

Outside of the aforementioned challenges, investigators faced no other major setbacks or problems in the conduction of this investigation.

Conclusion & Recommendations

In conclusion, this lab offered an in-depth introduction to memory forensics and forensic analysis by following a sound investigative process. Through the examination of a suspicious Windows XP memory image, investigators were able to utilize skills in the pursuit of following forensic procedures to simulate life-like incident response, including process enumeration, memory region inspection, DLL analysis, and executable extraction. Tools such Volatility 3, Volatility 2, and VirusTotal played a critical role in uncovering and validating the presence of malware within the system.

The learning objectives, such as identifying suspicious process behavior, examining memory artifacts, and understanding executable behavior in memory were successfully met. Investigators confirmed the presence of keylogger malware, while also examining its relationship with other processes, and inferred its likely method of execution and purpose.

This lab reinforced the importance of memory analysis in every facet of the investigative process. In the future, it is recommended that investigators continue to practice cross-version tool usage, become comfortable with manual inspection of memory structures, and maintain familiarity with malware behavior patterns. These skills are essential in real-world digital forensic investigations where speed, accuracy, and adaptability are critical to uncovering threats.

References

Volatility Foundation. (n.d.). *The Volatility Foundation*. <https://volatilityfoundation.org/>

Volatility Foundation. (n.d.). *Volatility 3 documentation (stable)*.

<https://volatility3.readthedocs.io/en/stable/>

VirusTotal. (n.d.). *Free online virus scan*. <https://www.virustotal.com/gui/home/upload>

Appendices

Contributions

Name(s)	Contributions to Investigation	Contributions to Report
Anthony Costa Jr.	Volatility 3 Analysis VAD & PeDump	Procedure & Analysis Challenges & Responses Executive Summary
Christopher Protheroe	Volatility 2 Analysis Process Enumeration Network Identification	Procedure & Analysis Conclusion & Recommends. Executive Summary